

Reference-Free Validation of Repetitive CFA Cancellation via an R-Phase Scramble Null, and a Cross-Subject Protocol for SeizeIT2

Working note for the CFA ear-EEG paper

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The problem this solves. There is no direct error feedback for the cancellation. The disturbance $d_r(t)$ is never observed in isolation ($y = s + d_r + n$), coherence to the ECG is diluted to near zero, and the FIR fit needs a clean d_r target that does not exist. So we cannot validate by comparing to a reference. This note builds a validation loop that needs *no* reference and *no* ground truth: it uses the one property that defines d_r — coherence with the true R-peak phase and nothing else — and measures everything as a contrast against a phase-*scrambled* null. Section 5 then scales this across SeizeIT2 subjects.

1 The scramble null is the only available handle

The single fact that separates d_r from all other terms is that it is phase-locked to the true R-peak instants $\{t_k\}$, while s and n are not. Averaging on the true peaks versus a phase-destroying scramble $\{t_k^{\text{scr}}\}$ gives

$$\hat{d}_{\text{true}}(p) = d_r(p) + \ell(p), \quad \hat{d}_{\text{scr}}(p) = 0 + \ell(p), \quad (1)$$

where $\ell(p)$ is the finite-record leakage of the asynchronous part, with $\|\ell\| \sim \text{SD}/\sqrt{N_{\text{eff}}}$. The scrambled template is therefore a *direct measurement of the leakage floor* — the denominator we otherwise could only derive. We have no ground-truth d_r , but we have a ground-truth *null*, and that is sufficient to build every metric below.

The R-phase-locked energy operator

For any signal $z(t)$ and any peak set P , define the template energy

$$\mathcal{E}_P[z] = \left\| \frac{1}{|P|} \sum_{k \in P} z(t_k + p/f_s) \right\|_2^2 \quad (\text{sum over template window } p = -n_{\text{pre}} \dots n_{\text{post}}). \quad (2)$$

This is “how much R-locked energy z carries under peak set P .” Every metric is a ratio of this operator under true vs. scrambled peaks. Because a single scramble is one draw, we repeat it B times ($B \sim 200\text{--}1000$ random scrambles) to obtain a null *distribution* $\{\mathcal{E}_{\text{scr}}^{(b)}\}$, which converts each ratio into a p -value rather than a bare number.

2 Three metrics (plus a safety guard)

Metric 1 — Locking ratio Q : was there anything to cancel?

On the raw measurement y ,

$$Q = \frac{\mathcal{E}_{\text{true}}[y]}{\mathcal{E}_{\text{scr}}[y]}, \quad p_Q = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\mathcal{E}_{\text{scr}}^{(b)}[y] \geq \mathcal{E}_{\text{true}}[y]]. \quad (3)$$

$Q \gg 1$ (with small p_Q) certifies a genuine R-locked deflection above the leakage floor: the template is real CFA/HEP, not averaged noise. This is the answer to “how do I know I am not cancelling nothing.” Since $\mathcal{E}_{\text{scr}}[y] = \text{SD}^2/N_{\text{eff}}$, this step also *measures* N_{eff} directly.

Metric 2 — Suppression S and residual locking Q' : was it removed?

After cancellation $\hat{s} = y - \hat{d}_{\text{true}}$, re-apply the operator to the *residual*:

$$S = 1 - \frac{\mathcal{E}_{\text{true}}[\hat{s}]}{\mathcal{E}_{\text{true}}[y]} \quad (\text{fraction of R-locked energy removed}), \quad (4)$$

$$Q' = \frac{\mathcal{E}_{\text{true}}[\hat{s}]}{\mathcal{E}_{\text{scr}}[\hat{s}]} \quad (\text{residual R-locked energy vs. floor}). \quad (5)$$

Complete cancellation drives $Q' \rightarrow 1$: the R-synced residual template becomes statistically indistinguishable from the async one. This is the formal version of “the R-sync and async residuals show the same trend.” The gap $Q' - 1$ is the *uncancelled R-locked fraction* — a self-contained error signal requiring no reference. Report S with an error bar from the null: the floor $\mathcal{E}_{\text{scr}}[y]$ and its spread give the noise band on S , so $S = 0$ falls inside the null and $S \rightarrow S_{\text{max}}$ at the floor.

Metric 3 — Neural-preservation guard: was $s(t)$ spared?

Metrics 1–2 live on the R-locked axis only. To confirm s was not damaged, compare band power (e.g. alpha 8–12 Hz) before vs. after in the async projection:

$$\Delta_\alpha = \frac{P_\alpha[\hat{s}] - P_\alpha[y]}{P_\alpha[y]}, \quad \text{require } |\Delta_\alpha| \lesssim 1/\sqrt{N_{\text{eff}}}. \quad (6)$$

A notch here ($\Delta_\alpha \ll 0$) signals over-cancellation — eaten HEP, or phase-scrambled leakage injected when N_{eff} is small. This guard keeps Metric 2 honest: forcing $Q' \rightarrow 1$ by brute force could damage s , and Δ_α catches it.

The headline number

S is the metric to report: reference-free, no coherence, no clean d_r , bounded and interpretable (0 = nothing removed, 1 = removed to the floor), with a null-derived error bar. Q certifies the target existed; Q'/S certify removal; Δ_α certifies safety.

3 How to scramble (this makes or breaks the null)

The null is valid only if the scramble destroys R-phase-locking while preserving *everything else* (beat count, interval distribution, spectral content). Three constructions, with different uses:

- **Circular time-shift** (primary): shift the whole peak train by a random offset $\gg$ one RR interval. Preserves the inter-beat-interval sequence *exactly* (including the RSA correlation structure), destroys absolute phase. Its floor therefore reflects the realistic N_{eff} you actually have. Use as the default null.
- **Interval shuffle**: permute RR intervals and rebuild peaks. Also destroys RSA correlation, so it *over*-estimates N_{eff} (ideal-independence floor). Use to bracket the best case, not to set the operating floor.
- **Phase-jitter**: add growing noise to true peaks for a *dose-response* curve (locking strength vs. jitter) — a clean robustness figure and a direct probe of timing sensitivity.

Report the circular-shift null as primary and build the B -sample distribution from it, so every Q, S, Q' carries a p -value.

4 The honest limit: R-locked $\neq$ cardiac-field

This framework validates removal of the *R-locked* component and *cannot* separate CFA from the heartbeat-evoked potential (HEP): both are phase-coherent with the R-peak, so both enter $\mathcal{E}_{\text{true}}$ and both are removed. The scramble certifies “R-locked energy removed,” not “cardiac-field energy removed.” Splitting those still needs the latency argument (CFA QRS-synchronous near $p = 0$; HEP at 200–600 ms) and the FIR-residual test — not the scramble. State S as R-locked suppression, and hand the CFA-vs-HEP axis to the other tool.

5 Cross-subject validation on SeizeIT2

The single-subject result (sub-001) makes Q and S *anecdotes*. The generalization claim the paper needs is the *distribution* of Q and S across a representative cohort. This is also the correct scope for a conference paper: a stratified subset, not all 2850 recordings.

5.1 The confound that matters most: ECG→EEG cross-device timing

SeizeIT2 records EEG on the behind-the-ear SD module and ECG on a *separate* chest SD module, each with its own clock; the dataset paper documents inter-device clock drift and non-exact sampling corrected by cross-correlation against full-scalp EEG. For a *timing-anchored* method this is not incidental: any residual ECG→EEG misalignment or drift is *indistinguishable from weak locking* — it directly lowers Q and caps S . Before trusting any per-subject number:

1. Verify (or re-estimate) ECG–EEG sample alignment per recording; correct residual drift.
2. Use the null as automatic QC: a recording with broken alignment shows $Q \approx 1$ (no locking) even though CFA is physically present. Flag low- Q recordings and inspect *alignment* before concluding “weak CFA.” The scramble test doubles as a timing-integrity check across the cohort — a genuinely useful by-product.

5.2 Stratification

CFA magnitude depends on electrode geometry and population. Stratify and report within strata, do not pool blindly:

- **Center** (5 EMUs): equipment/montage practice varies.

- **Montage/channel:** same-side (left/right) vs. cross-head vs. generalized lateral. CFA scales with proximity to neck vasculature, so report Q, S per channel type.
- **Age:** pediatric (Leuven only) vs. adult — different HR, head size, CFA amplitude.

5.3 Segment selection (state the rule; avoid cherry-picking)

- **Use interictal/background segments** for clean CFA characterization: during seizures, $s(t)$ is large and abnormal, motion inflates $n(t)$, and HR/rhythm shifts degrade R-peak detection — all of which depress Q for reasons unrelated to the canceller. Characterize on resting windows; optionally report ictal separately as a stress case.
- **Exclude the $\sim 3\%$ of recordings lacking ECG** (no reference timing possible).
- **Fixed-length window per recording** (e.g. 10–20 min interictal, matching the $N_r \sim 700$ -beat regime of the paper) chosen by a rule (first qualifying clean window), not by CFA visibility. State the rule explicitly.

5.4 Per-recording pipeline

For each selected recording: (1) detect R-peaks on ECG with the min/max-PI tracker + arrhythmia supervisor; (2) build $\hat{d}_{\text{true}}$ and B circular-shift $\hat{d}_{\text{scr}}$; (3) compute Q, p_Q and N_{eff} ; (4) cancel, compute S and Q' ; (5) compute Δ_α . Emit one row per (subject, channel) with these five numbers plus beat count and a timing-integrity flag.

5.5 Aggregation and reporting

- **Distributions, not means:** box/violin plots of Q and S across subjects, split by stratum. This *is* the generalization figure.
- **Subject as the unit:** multiple recordings/channels per subject are correlated; aggregate to a per-subject value (median) for the headline, and note within-subject spread separately (a mixed-effects framing if you want rigor).
- **Report the failure fraction:** proportion of recordings with p_Q not significant. Interpreted correctly (§5.1), a low- Q recording is usually a timing/alignment failure, not absent CFA — disclose both readings.
- **Safety across cohort:** distribution of Δ_α ; it should straddle zero within $\pm 1/\sqrt{N_{\text{eff}}}$, demonstrating preservation is not subject-specific luck.

5.6 Scale

For a conference paper, 20–40 subjects spanning the five centers, both age groups, and each montage type, with one interictal segment each, is sufficient and defensible. A few subjects with multiple segments establish within-subject stability. Running all 125×2850 is journal-scale effort misallocated to a 4-page result; the stratified subset delivers the distribution claim at a fraction of the cost.

Bottom line. The scramble null converts “no error feedback” into a complete, reference-free loop: Q says the disturbance was there, S/Q' say it was removed to a measured floor, Δ_α says s

was spared — all without ground truth. Across SeizeIT2, the same null doubles as a cross-device timing-integrity check, and the distribution of Q, S over a stratified 20–40-subject subset is the generalization result the paper currently lacks.